Graph Theory & Networks (AI HL)

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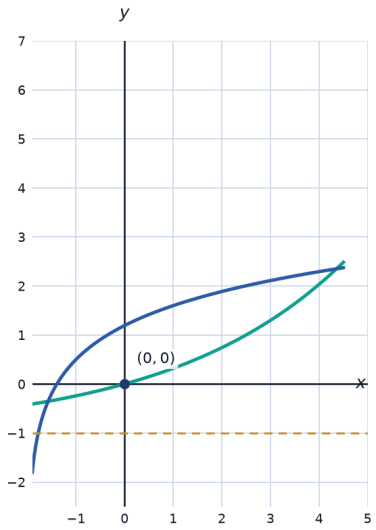
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Variables x and y satisfy $y = ax^b$. A graph of $\ln y$ against $\ln x$ is a straight line passing through $(0, 1.20)$ and $(4, 6.80)$. Find the values of the constants a and b to 3 significant figures.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Given $\ln y = 2\ln x + \ln 3$, express y in terms of x. Hence sketch the graph.



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A student calculates $\int_1^3 \frac{4}{2x+1}\,dx$ by rounding $\ln 7 \approx 1.95$ and $\ln 3 \approx 1.10$ at the start, obtaining $2(1.95 - 1.10) = 1.70$. State the exact value of the integral and calculate the absolute error introduced by premature rounding.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Level 2 · Medium Use the method of cylindrical shells to find the volume generated by revolving the region bounded by $y = 3x - x^2$ and $y = 0$ about the y-axis.

5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the vertex, focus, and directrix of the parabola given by the general equation $y^2 + 8x - 4y + 28 = 0$.

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A 3 kg particle on a rough surface, $\mu = 0.4$. Horizontal force F N. (a) Find the limiting friction. (b) Acceleration when $F = 20$ N. (c) Time to travel 5 m from rest.

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7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 2 · MediumFind the derivative $\frac{dy}{dx}$ for the function: $y = \int_{\cos x}^{x^3} \sqrt{1 - t^2}\, dt$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 1 · Foundations Find the area of the region bounded by $y = x^2$ and $y = 2x - x^2$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the set of values of x satisfying the inequality $2x^2 - 7x + 3 \leq 0$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The equation $x^3 - x - 1 = 0$ has a root near $x = 1.3$. (a) Show that the iterative formula $x_{n+1} = (x_n + 1)^{1/3}$ converges by considering $g'(x)$. (b) Starting with $x_1 = 1.3$, calculate x_2 and x_3 to 4 decimal places.

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Level 1 · Foundations Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = abla(x^3y + 2z)$ along any curve C from $(0, 0, 1)$ to $(2, 1, 3)$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Level 1 · FoundationsFind the exact natural domain of the function $g(x) = \sqrt{\frac{x-1}{x+3}} - \ln(5 - x)$ in interval notation.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the 15th term and the sum of the first 20 terms of the arithmetic progression $4, 9, 14, 19, \ldots$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 1 · FoundationsFind $h'(2)$ given the accumulation function $h(x) = \int_0^{x^2} \frac{t}{t+1}\, dt$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

A student computes the volume of a cylinder using $r = 3.2$ cm and $h = 7.85$ cm. The raw calculator display shows $252.518...$. How should the final answer be reported according to Cambridge AS Level guidelines?

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16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

A radioactive isotope has a decay constant $k = 0.035\ \text{year}^{-1}$. (a) Find its half-life to the nearest tenth of a year. (b) If an initial sample contains $250\ \text{mg}$, how much remains after $40\ \text{years}$?

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Matrices (AI HL)

$f(x) = cx^2$ for $0 \le x \le 3$, zero otherwise. Find c, $E(X)$, and $P(1 \le X \le 2)$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.03 Markov Chains (AI HL)

A manufacturer claims that the mean breaking strength of cables is at least 80 kN. A sample of 16 cables gives $\bar{x} = 77.5$ kN with population SD 5 kN. Test this claim at 1% significance.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Proof by Mathematical Induction

A discrete random variable Y has $E(Y) = 5$ and $\text{Var}(Y) = 4$. Calculate $E(Y^2)$ and $\text{Var}(10 - 3Y)$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

Level 1 · Foundations Find $\frac{dy}{dx}$ in terms of t for the parametric curve $x = 2\cos t$, $y = 5\sin t$ for $t \in (0, \pi)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What does the Handshaking Lemma state for any undirected graph with V vertices and E edges?

a. The sum of the degrees of all vertices equals twice the number of edges: $\sum \deg(v) = 2E$

b. The sum of the degrees equals E

c. The number of vertices with odd degree is always odd

d. Every vertex must have degree at least 2

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A graph has 6 vertices, each having degree 3. How many edges does the graph have?

a. 9

b. 18

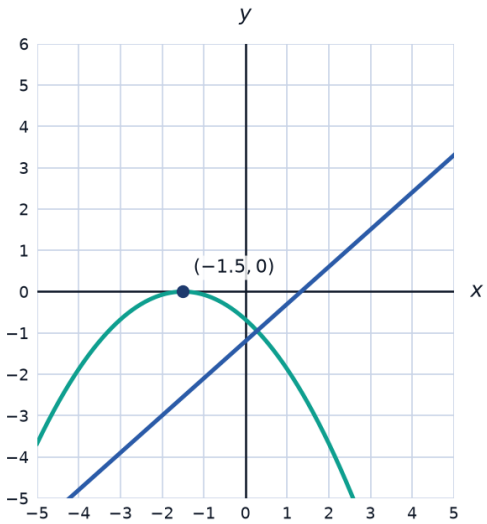
c. 6

d. 12

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Why can a graph never have an odd number of vertices of odd degree?



a. Because $\sum \deg(v) = 2E$ is always even, the sum of odd degrees must be even, forcing an even count of odd vertices

b. Because edges cannot cross each other

c. By Euler's formula $V - E + F = 2$

d. Because every vertex must have at least one adjacent edge

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What condition is both necessary and sufficient for a connected graph to contain an Eulerian circuit (a closed trail visiting every edge exactly once)?

a. Every vertex has an even degree

b. Exactly two vertices have odd degree

c. The graph contains no cycles

d. The graph is a complete bipartite graph

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If a connected graph has exactly two vertices of odd degree, what type of Eulerian walk does it contain?

a. An Eulerian trail (open path visiting every edge once) that starts at one odd vertex and ends at the other

b. An Eulerian circuit returning to the start

c. No Eulerian trail or circuit can exist

d. A Hamiltonian cycle

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the difference between an Eulerian circuit and a Hamiltonian cycle?

a. An Eulerian circuit visits every edge once; a Hamiltonian cycle visits every vertex once

b. An Eulerian circuit visits every vertex once; a Hamiltonian cycle visits every edge once

c. They are identical concepts in graph theory

d. Hamiltonian cycles apply only to directed trees

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a connected weighted graph with V vertices, how many edges does any spanning tree contain?

a. $V - 1$

b. V

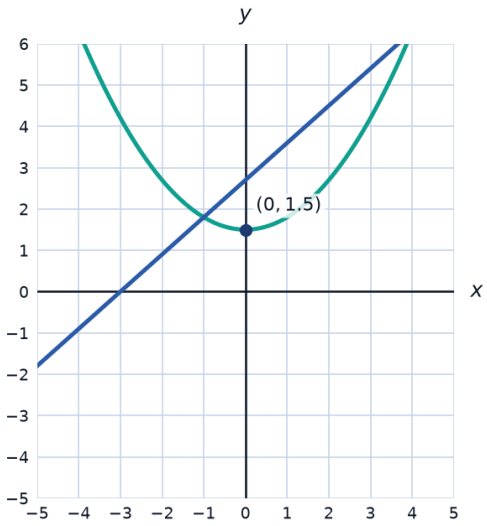
c. $V + 1$

d. $\frac{V(V-1)}{2}$

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

How does Kruskal's algorithm select edges to construct a Minimum Spanning Tree (MST)?



a. Sorts all edges by weight and greedily adds the smallest available edge that does not create a cycle

b. Grows a single tree by continuously adding the closest vertex to the current tree

c. Removes the largest cycle edges one by one until only a tree remains

d. Visits vertices in alphabetical order

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

How does Prim's algorithm differ conceptually from Kruskal's algorithm?

a. Prim grows a single connected tree from an arbitrary starting vertex, whereas Kruskal maintains a forest of components until unified

b. Prim works only on unweighted graphs

c. Kruskal finds the longest path rather than minimum tree

d. Prim can only be applied to planar graphs

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the purpose of Dijkstra's algorithm?

a. To find the shortest paths from a single source vertex to all other vertices in a network with non-negative edge weights

b. To find a minimum spanning tree

c. To determine if a graph has an Eulerian circuit

d. To find the maximum flow through a network

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Why does Dijkstra's algorithm fail if a network contains negative edge weights?

a. A greedy choice once finalized could be improved later by a negative edge, invalidating the optimality assumption

b. Distances cannot be negative by definition

c. The algorithm enters an infinite cycle immediately

d. Square roots of negative weights cannot be computed

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In a planar graph, if $V = 6$ and $E = 9$, how many faces F does the graph enclose according to Euler's formula $V - E + F = 2$?

a. 5

b. 3

c. 4

d. 6

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

If $f''(x) > 0$ on an interval, the graph of f is:

a. concave up

b. decreasing

c. concave down

d. linear

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Given $f(x) = (x - 2)^2(x + 1)$, does the graph touch or cross the x-axis at $x = 2$?

a. The graph does not intersect the x-axis there

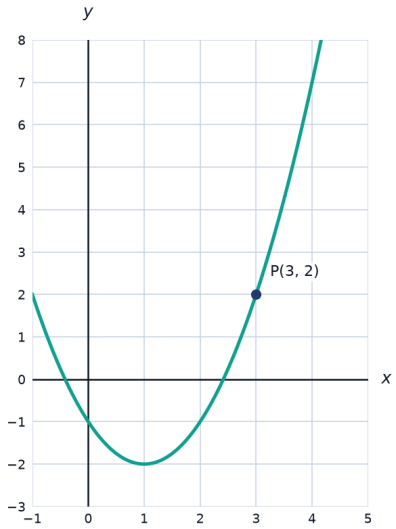
b. Not enough information

c. Touches (does not cross)

d. Crosses

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The figure shows the graph of the function $f(x) = x^2 - 2x - 1$ and the point $P(3, 2)$ on the curve. What is the slope of the tangent line to the graph of f at point P?



a. 4

b. 2

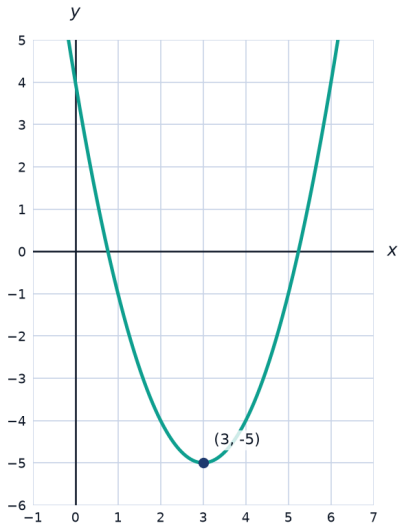
c. 3

d. 6

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36 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

The graph of the quadratic function $f(x) = x^2 - 6x + k$ is shown. The vertex of the parabola is at the point $(3, -5)$. Find the value of the parameter k.



a. -5

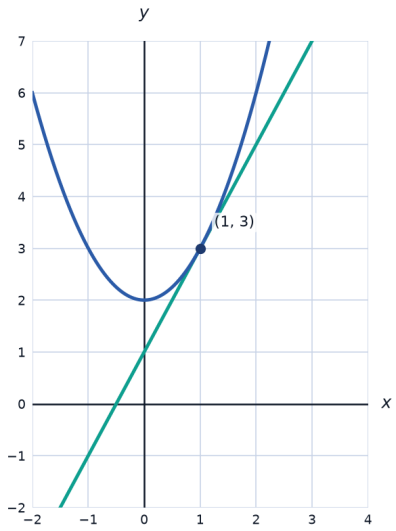
b. 4

c. 9

d. 0

37 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

The graph of a differentiable function $f(x)$ is shown below. The function has a tangent line at $x = 1$ with equation $y = 2x + c$. If $f(1) = 3$, find the value of c.



a. 1

b. 2

c. 3

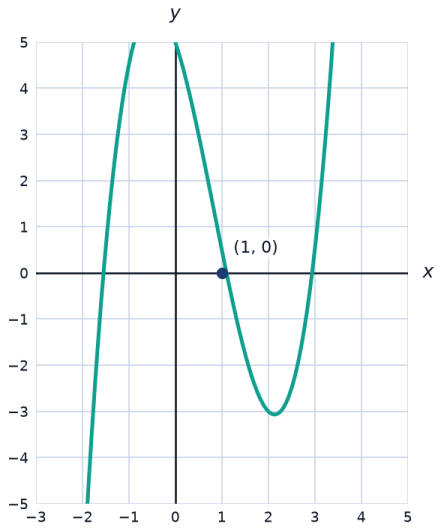
d. 4

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 6.03 Markov Chains (AI HL)**

Consider the graph of the function $f(x)$ shown below. Which of the following points lies on the graph of $f(x)$?



a. $(-2, 3)$

b. $(0, -1)$

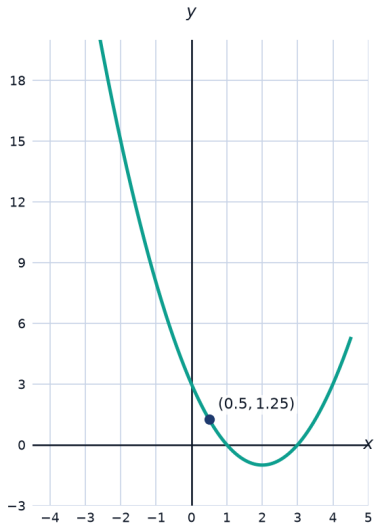
c. $(1, 0)$

d. $(3, 4)$

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Proof by Mathematical Induction

The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. The function has a local minimum. What are the coordinates of the local minimum?



a. (2, 1)

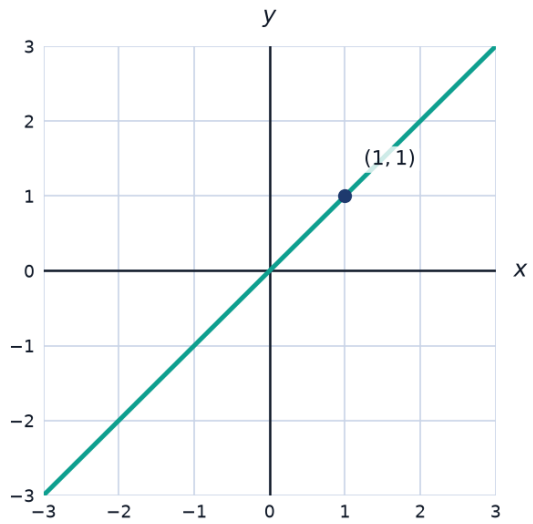
b. (2, -1)

c. (1, 0)

d. (3, 0)

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

Consider the graph of the function $f(x)$ shown below. What is the limit of $f(x)$ as x approaches 1?



a. 1

b. 0

c. 2

d. Does not exist

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Consider the function $f(x)$ whose graph is given below. The function is defined piecewise. For $x \leq 2$, $f(x) = ax + b$. For $x > 2$, $f(x) = cx + d$. The graph of $f(x)$ passes through the point $(0, 3)$. The limit of $f(x)$ as x approaches 2 exists. Find the value of a.

a. -1

b. 1

c. 2

d. -2

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.04 Partial Fractions

Consider the rational function $f(x) = \frac{ax^2 - 4x + 1}{x^2 - bx + 2}$. The graph of $y = f(x)$ has a vertical asymptote at $x = 1$ and a horizontal asymptote at $y = 2$. If $f(0) = \frac{1}{2}$, find the value of a.

a. 1

b. 2

c. 3

d. 4

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Consider the rational function $f(x) = \frac{x^2 - ax + 6}{x - b}$. If the graph of $f(x)$ has a removable discontinuity at $x = 2$ and a vertical asymptote at $x = 3$, determine the values of a and b.

a. a = 5, b = 3

b. a = -5, b = 3

c. a = 5, b = 2

d. a = -5, b = 2

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

The diagram shows the graph of the function $y = f(x)$. The graph is transformed by a sequence of operations. The resulting graph, $y = g(x)$, is also shown. The function $f(x)$ is given by $f(x) = ax^2 + bx + c$, where $a, b,$ and c are constants. The vertex of the graph of $y = f(x)$ is at $(2, -1)$, and it passes through the point $(4, 3)$. The graph of $y = g(x)$ is obtained by translating the graph of $y = f(x)$ by $(-1\ 3)$ followed by a reflection in the y-axis. Find the values of $a, b,$ and c.

a. $a = 1, b = -4, c = 3$

b. $a = 1, b = 4, c = -5$

c. $a = 2, b = -8, c = 7$

d. $a = -1, b = 4, c = -5$

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 3D Vectors & the Cross Product

The graph of the function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of the parameter a.

a. 1

b. 2

c. 3

d. 4

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